

Nickolas Yanez

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EDUCATION

The University of Texas at Austin

B.S. Aerospace Engineering Honors, Minor in Robotics

Austin, TX

May 2027

- **GPA:** 3.82/4.00

EXPERIENCE

Team Lead — Active Roll Control

Apr. 2026 – Present

Lonehorn Rocketry Association - Advanced Research

Austin, TX

- Led transition from actuated fins to four-canard roll control to address packaging and serviceability limitations
- Designed a 3.1-in subscale vehicle with 1.38-cal stability and sized actuators for 0.141 N·m design torque
- Designed the first iteration of a four-servo canard bay in SolidWorks for actuator packaging and vehicle integration
- Developed a MATLAB roll-control simulation coupling OpenRocket trajectories, CAD-derived inertia, canard aerodynamics, and servo dynamics

Mechanical Member — Air Brakes

Aug. 2025 – Apr. 2026

Lonehorn Rocketry Association- Advanced Research

Austin, TX

- Contributed to servo-actuated air brakes targeting apogee reduction from 12,000 ft to 10,000 ft
- Analyzed deployment aerodynamics across the expected Mach range to estimate drag increase
- Performed FEA of deployed brake surfaces to evaluate stress and deformation under aerodynamic loading

Vehicle Integration Member - Flight Hardware

Aug. 2024 – Aug. 2025

Lonehorn Rocketry Association- Flight Systems

Austin, TX

- Fabricated and integrated composite airframes, fins, bonded structures, and recovery hardware
- Analyzed competition vehicle stability in OpenRocket, verifying a 2.75-cal stability margin
- Supported vehicle integration, recovery preparation, and launch-readiness checks during test flights
- Selected as 1 of 17 members to support flight operations at the 2025 Spaceport America Cup

PROJECTS

Quadrotor Dynamics Simulation & Control | *Aerial Robotics*

Spring 2026

- Built a MATLAB quadrotor simulation with 6-DOF dynamics, motor modeling, and aerodynamic drag
- Implemented closed-loop position and attitude control with simulated GNSS/IMU sensing and UKF state estimation
- Implemented stereo 3D localization in OpenCV using RANSAC outlier rejection and least-squares triangulation

3-DOF Robotic Leg | *Gateway to Robotics*

Fall 2025

- Fabricated a 3-DOF robotic leg using 3D-printed parts, laser-cut hardware, and Dynamixel servos
- Implemented forward and inverse kinematics in Python using homogeneous transformations and Newton's method
- Integrated hardware with PyBullet for bidirectional simulation control and experimentally evaluated PD response

Satellite Targeting | *Spacecraft Dynamics*

Fall 2025

- Implemented Skopinski-Johnson orbital targeting in MATLAB for a multi-revolution overflight mission
- Propagated the targeted orbit with J2 perturbations and Earth rotation to validate ground tracks
- Targeted an Austin overflight within 0.213° latitude and 0.162° longitude of the desired location

TECHNICAL SKILLS

Programming: MATLAB, Python, C++ | **Tools:** SolidWorks, Ansys, OpenRocket, ROS, OpenCV, LabVIEW, Git, Linux

Coursework: Attitude Dynamics, Flight Dynamics, Aerial Robotics, Feedback Control Systems, Spacecraft Dynamics

Honors: Hans Mark Engineering Honors Scholar, CH Foundation Scholar